

HMO Licence Retention and Rental Supply in St Andrews

Administrative-data construction, property matching, regression analysis, and historical web scraping

Project objective

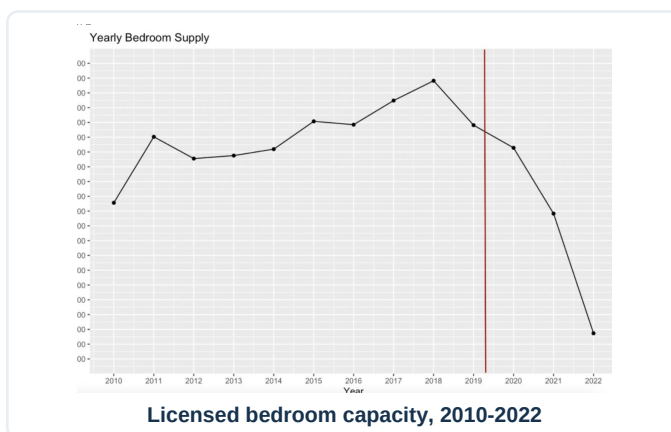
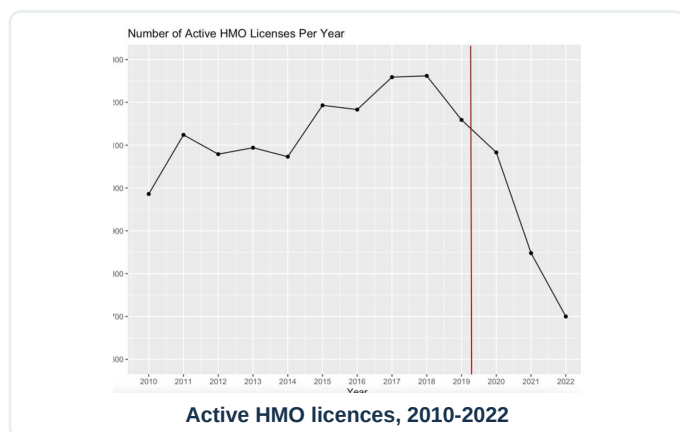
Measure changes in the supply and retention of HMO-licensed properties surrounding Fife Council's 2019 freeze on new HMO licences, and examine historical rental-price evidence from archived listings.

Data and methodology

Converted Fife Council's public HMO register from PDF to Excel and used R to clean, standardize, deduplicate, and match property-level records from 2004-2022. Constructed annual active-licence and licensed-bedroom measures, estimated licence-retention models for 2015-2022, and repeated the analysis for 2014-2021. Python, Selenium, Beautiful Soup, and the Wayback Machine were used to collect historical rental listings.

Key findings

- Constructed a cleaned longitudinal database containing 4,918 HMO licence records.
- Active HMO licences declined from 1,262 in 2018 to 700 in 2022, a 44.5% decrease.
- Licensed bedroom capacity declined from 7,764 in 2018 to 4,347 in 2022, a 44.0% decrease.
- Five-occupant and six-plus-occupant properties had materially higher licence-retention rates than three-occupant properties.
- Conservation-area status was not statistically significant in the main or alternate-period model.
- The historical scraper collected about 1,400 raw listings and produced 478 relevant St Andrews records.



Interpretation and limitations

The analysis documents a substantial contraction in the observed stock of active HMO licences and licensed bedroom capacity surrounding the policy period. Because the design is observational and lacks an untreated comparison group, the decline should not be interpreted as a fully causal estimate of the 2019 freeze. Archived rental listings are incomplete, nonrandom, and uneven across years; those results are exploratory. The Ayton House analysis remains in the repository, although its original input file is no longer available for rerunning.

Tools: R, Python, Selenium, Beautiful Soup, pandas, Excel, regression modeling, record linkage, and data visualization.